MATH3881/5881: Statistical Machine Learning Theory

Week 3: Theory Tutorial: Modified Dropout as Ensemble Averaging

Sarat Moka *UNSW, Sydney*

A single short problem to set up the Practice Tutorial. It draws on the last section of the Week 2 lecture notes (*Regularisation*, specifically the dropout subsection) and is the mathematical content behind Problem 2 of the Week 3 Practice Tutorial. Aim to finish it within the first 10 minutes of the tutorial so you have plenty of time for the notebook. A separate solutions document is provided a week later.

Problem: Modified dropout is unbiased gradient descent in expectation

Let $C : \mathbb{R}^d \to \mathbb{R}$ be a smooth loss function and $\theta \in \mathbb{R}^d$ the current parameter vector. Consider the following *per-weight modified dropout*: with keep-probability $q \in (0, 1]$, sample an i.i.d. Bernoulli mask $m \in \{0, 1\}^d$ *on parameters* (one Bernoulli per weight), $m_i \sim \mathrm{Bern}(q)$, and form the modified gradient

$$\tilde{g} \;=\; \tfrac{1}{q}\, m \odot \nabla C(\theta),$$

where $\odot$ is the elementwise (Hadamard) product. One step is then $\theta \mapsto \theta - \alpha\tilde{g}$. The mask is independent of θ and is freshly sampled at every step. (Standard dropout, recalled in Week 2 §6, masks *activations* in the forward pass; the per-weight scheme studied here is a simpler variant that makes the BCD interpretation exact.)

(a) Compute $\mathbb{E}[\tilde{g}]$, where the expectation is over the mask only. Conclude that one modified-dropout step is, *in expectation*, one standard gradient-descent step from Week 1 with step size α.

(b) Now consider the same scheme *without* the $1/q$ rescaling, that is, $\tilde{g}' = m \odot \nabla C(\theta)$. Show that $\mathbb{E}[\tilde{g}'] = q\,\nabla C(\theta)$. Conclude in one sentence why PyTorch's `nn.Dropout` uses the $1/q$ factor and not the raw mask.

(c) *Connection to the Practice Tutorial.* Consider a *linear* model $f(\theta; x) = \theta^\top x$. Show that

$$f(\theta; x) \;=\; \mathbb{E}_m\Big[\, f\Big(\tfrac{1}{q}\, m \odot \theta;\, x\Big)\Big],$$

exactly, where the expectation is over the dropout mask m.